

a Nobel laureate in every paragraph. Bell, however, is the exception that proves the rule. He died in 1990, and since Nobel Prizes are not awarded posthumously, he never received the honour. Although Bell never received a Nobel Prize, his theorem paved the way for others to receive science's highest honour and helped settle one of the great philosophical disputes of modern physics.

The four maximally entangled two-qubit states are named the Bell states in his honour. They are shown in Figure 3, written in the two notations this series has used since the first article: Dirac notation, the bra-ket notation named after the physicist and Nobel laureate Paul Dirac, and the vector form, which lists the four amplitudes in the order $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$, the superscript T marking the printed row as a column vector. Notice their structure: two of them, conventionally written $|\Phi^+\rangle$ and $|\Phi^-\rangle$, are superpositions of $|00\rangle$ and $|11\rangle$, and the other two, $|\Psi^+\rangle$ and $|\Psi^-\rangle$, are superpositions of $|01\rangle$ and $|10\rangle$. Within each pair, the two states differ only by a minus sign—which is to say, only by a phase. The state we have been producing since the previous article is $|\Phi^+\rangle$, obtained from $|00\rangle$ by a Hadamard gate followed by a CNOT gate.

Name	Dirac notation	Vector form
$ \Phi^+\rangle$	$\frac{ 00\rangle + 11\rangle}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}[1 \ 0 \ 0 \ 1]^T$
$ \Phi^-\rangle$	$\frac{ 00\rangle - 11\rangle}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}[1 \ 0 \ 0 \ -1]^T$
$ \Psi^+\rangle$	$\frac{ 01\rangle + 10\rangle}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}[0 \ 1 \ 1 \ 0]^T$
$ \Psi^-\rangle$	$\frac{ 01\rangle - 10\rangle}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}[0 \ 1 \ -1 \ 0]^T$

Figure 3: The four Bell states

Three properties make these four states special. First, they are maximally entangled: measuring either qubit alone yields 0 or 1 with equal probability, so an individual qubit of a Bell pair carries no information whatsoever, while the pair together is perfectly correlated. All the information lives in the relationship, not in the parts. Second, they are mutually orthogonal and form a complete basis for two qubits, called the Bell basis. Any two-qubit state can be written as a combination of these four, just as any single-qubit state can be written in terms of $|0\rangle$ and $|1\rangle$. Third—and this is the property we shall exploit shortly—all four can be reached from $|\Phi^+\rangle$ by operating on just one of the two qubits, leaving the other untouched.

Before we write any more code, a small but important warning about notation. Qiskit prints its bit strings with the highest-numbered qubit on the left. In a result such as '01', the right-hand character is the outcome of qubit 0 (1 in this

case) and the left-hand character is the outcome of qubit 1 (0 in this case). Until now nothing has depended on this. Our entangled results were 00 and 11, which read the same in either direction, and the stray 01 and 10 counts in Figure 1 were simply counted together as errors. From this point onwards, 01 and 10 are different answers and the ordering genuinely matters. This convention has confused more beginners than any quantum-mechanical subtlety, so it is worth committing to memory.

Now let us understand, at the mathematical level, what entanglement means. A two-qubit state is called separable, or unentangled, if it can be written as the product of two independent single-qubit states. Suppose $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ were separable. Then we could write it as $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$, which expands to $\alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$. Comparing coefficients, we need $\alpha\gamma = 1/\sqrt{2}$ and $\beta\delta = 1/\sqrt{2}$, so none of the four numbers may be zero. But we also need $\alpha\delta = 0$ and $\beta\gamma = 0$, which forces at least one of them to be zero. The two requirements contradict each other, so no such factorisation exists. That short argument is the whole definition: an entangled state is one that cannot be described by describing its parts separately. There is no mysterious influence travelling between the qubits. There is simply no longer any such thing as 'the state of qubit 0'.

Figure 3 tells us what the four states are, but not how to make them. All four begin with the same Hadamard and CNOT pair that produced $|\Phi^+\rangle$ in the previous article; what differs is the operation applied afterwards, and it acts on qubit 1 alone. Z turns $|\Phi^+\rangle$ into $|\Phi^-\rangle$, X turns it into $|\Psi^+\rangle$, and X followed by Z turns it into $|\Psi^-\rangle$. That is the third property above, expressed as a recipe: the four Bell states differ only by an operation on a single member of the pair.

Now look at what separates $|\Phi^+\rangle$ from $|\Phi^-\rangle$. It is a single minus sign, and by now that should be setting off an alarm bell. In the third article of this series we met exactly this situation with a single qubit: the Pauli-Z gate appeared to do nothing at all, because a phase cannot be seen in the measurement statistics. Does the same disappointment await us here?

Telling the Bell states apart

Let us find out. The program `qiskit11.py`, shown below, prepares all four Bell states in turn and measures each of them in the computational basis.

```
1. from qiskit import QuantumCircuit, transpile
2. from qiskit_aer import AerSimulator

3. bell_states = {"Phi+": "", "Phi-": "Z", "Psi+": "X",
                 "Psi-": "XZ"}
4. sim = AerSimulator( )
```